Vacuum 236 (2025) 114137

Contents lists available at ScienceDirect

Vacuum

journal homepage: www.elsevier.com/locate/vacuum

Short communication

Study on cavitation flow field of cavitation nozzle and strengthening mechanism of cavitation jet of 7075 aluminum alloy

Xiujie Yue [a], Ping Zhang [b,*], Yajie Sun [b], Xiaomin Jiang [b], Youqiang Wang [a,**]

[a] College of Intelligent Manufacturing, Qingdao University of Technology, Qingdao, 266520, China
[b] College of Mechanical and Power Engineering, Guangdong Ocean University, Zhanjiang, China

ARTICLE INFO

Handling Editor: Luigia Sabbatini

Keywords:
Nozzle structure
Jet velocity
Jet pressure
Surface enhancement
Residual stress

ABSTRACT

This study aims to explore the influence of the divergence angle of a nozzle on the flow field structure of a water jet. Based on the traditional conical nozzle structure, two types of nozzles were designed: angular nozzles with divergence angles ranging from 0° to 30° and abrupt expansion nozzles with divergence angles ranging from 60° to 180°. A numerical model was established to analyze the relationship between the nozzle divergence angle and the nozzle exit velocity. The model was validated using Fluent simulations, and the impact of the nozzle divergence angle on the water jet flow field structure was investigated.The research findings indicate that the numerical model provides a good fit to the nozzle exit jet velocity with a maximum error of only 14.1 % when compared to the simulation model. This suggests that the model has practical implications for predicting the exit jet velocity of conical nozzles, allowing for rapid assessment of jet velocities under different nozzle exit divergence angles. Furthermore, as the divergence angle increases, the nozzle exit jet velocity exhibits a trend of initially decreasing, then increasing, and finally decreasing again. When the nozzle divergence angle is 60°, the exit jet velocity is the lowest, measuring 171 m/s, while an increase in the divergence angle to 90° results in a 7 % velocity increase. At a 60° divergence angle, the axial velocity decreases most rapidly, leading to jet deflection. Under the same operating conditions, a 90° divergence angle nozzle yields the highest exit dynamic pressure at 11.646 MPa, an increase of 19 % compared to the 60° divergence angle nozzle. Additionally, at the water jet exit, dynamic pressure exceeding 2 MPa is considered effective impact pressure. At a 60° divergence angle, the effective impact diameter is approximately six times larger than that at 30°, with a 2–3 mm greater effective impact diameter compared to other divergence angles, making it more suitable for surface strengthening operations on metals.Finally, different divergence angle nozzle exit velocities, obtained through fitting, were used to validate the residual stress in 7075 aluminum alloy after water jet impact using ABAQUS simulations and experiments. After water jet impact, 7075 aluminum alloy exhibits primarily residual compressive stress, and the distribution of residual stress resembles a "spoon." When comparing experimental and simulation values, residual stress changes along the depth direction are generally consistent, with errors ranging from 3.7 % to 39.6 %.

Nomenclature

C_d	Flow coefficient	d_ε、d_δ	The diameter of the fluid inside and outside the nozzle
d_0	Nozzle outlet diameter	v_β	Actual fluid velocity
δ^*	Boundary layer displacement thickness	q_α	Jet flow
K_p	Pressure gradient parameter	Q_β	Actual flow rate

(*continued on next column*)

(*continued*)

μ	Viscosity of water movement	Q_α	Theoretical flow
S_c	Laminar layer thickness	C_V	Speed factor
L_1	Length of convergence section	a	Natural Constants
L_2	Length of cylindrical section	τ	Volumetric micro elements

(*continued on next page*)

* Corresponding author.
** Corresponding author.
E-mail addresses: zpqdhhxy@126.com (P. Zhang), xjyue_ytu@126.com (Y. Wang).

https://doi.org/10.1016/j.vacuum.2025.114137
Received 13 October 2024; Received in revised form 20 December 2024; Accepted 14 February 2025
Available online 20 February 2025
0042-207X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(*continued*)

L_3	Length of diffusion section	p_d	Water jet dynamic pressure
R_0	Diameter of nozzle inlet	p_0	Water jet inlet pressure
R_1	Diameter of cylindrical section	ρ	Density of water
R_2	Diameter of nozzle outlet	E	Kinetic energy of water jet
α	Convergence angle	f_x、f_y、f_z	Volume force per unit mass of fluid in the x, y and z directions
β	Diffusion angle	G_k	Turbulent energy due to mean velocity gradient
p_e、p_δ	Nozzle internal and external static pressure	G_b	Turbulent energy due to buoyancy effects
v_e、v_δ	Average velocity of fluid inside and outside the nozzle	Y_M	Effect of pulsating expansion of compressible velocity turbulence on the total dissipation rate
d_x	Diameter of any section of nozzle	μ_t	Turbulent viscosity coefficient

1. Introduction

With the modernization and automation of industry, there is an increasing demand for improved mechanical, physical, and chemical properties of metal components in service environments [1–3]. These properties are highly dependent on the integrity of the material's surface. To overcome the shortcomings of metal materials such as poor corrosion resistance and susceptibility to fatigue fractures [4,5], surface modification plays a crucial role in expanding their application range and prolonging their lifespan. Surface modification techniques involve methods that enhance the physical and chemical properties of components while keeping the basic chemical composition of the base material relatively unchanged [6,7]. Among these techniques, water jet technology has gained widespread use due to its environmentally friendly and versatile nature [8], making it suitable for operation in confined spaces, complex environments, and harsh conditions.

The nozzle is a critical component in water jet enhancement systems, significantly influencing the dynamic performance of water jets and the distribution of internal flow fields [9–11]. The shape and dimensions of the water jet nozzle's channel have a decisive impact on its performance, resulting in changes in the internal fluid and consequently affecting the flow characteristics at the nozzle exit [12,13], directly impacting the nozzle's impingement performance. Therefore, to enhance the efficiency of water jet strengthening, it is necessary to have nozzles with excellent performance and structure.

In studying water jet flow fields, Liu et al. [14] analyzed the flow field of intermittent pulsed water jets in hard rock fragmentation, capturing the evolution of individual pulses using high-speed cameras and analyzing them through numerical simulations. Their research revealed that the velocity at the nozzle inlet is lower than the nozzle exit velocity, but the former has higher turbulence energy. The jet forms a U-shaped structure upon contact with the target material, affecting the peak impingement pressure, duration, and distribution on the target. Han et al. [15] used numerical simulations to study the impact of water jets on different curved contact surfaces, revealing flow field and coal damage characteristics. They found that the water jet impingement flow field is primarily divided into three regions: velocity concentration, divergent flow, and wall return flow. Moreover, as the jet water pressure increases, the distribution characteristics of the flow field become more pronounced. Near n = 0.8, the ratio of the peak velocity to the axis decreased to a minimum, indicating the maximum kinetic energy loss during the impingement of the water jet on the contact surface. When water jets of different pressures impact contact surfaces, the peak velocity drops by two-thirds before and after impact, with approximately 89 % of the kinetic energy lost during the impingement process. This clarifies the flow field and coal damage characteristics of water jet impingement on contact surfaces with different curvatures. Therefore, flow field analysis for water jets mainly focuses on dynamic pressure and peak velocity, but the influence of nozzle structure should not be ignored [16,17].

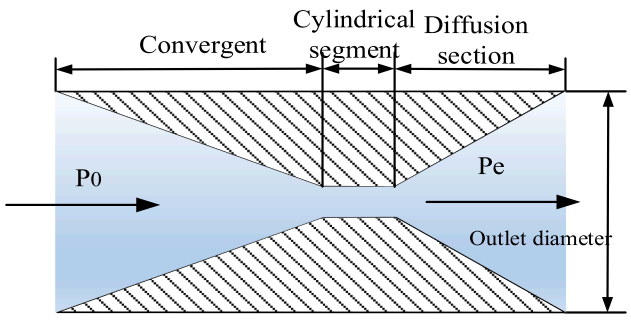


Fig. 1. Structural schematic of the converging and expanding nozzle.

The structural design of water jet nozzles directly affects the dynamic characteristics and enhancement effect of the jet. Optimizing the geometric parameters of the nozzle to improve jet performance has become a key research direction [18–20].Regarding the impact of nozzle structure on water jet characteristics, Chen et al. [21] proposed an optimization platform for cavitation nozzles, taking into account geometric parameters of the nozzles and using CFD technology to develop a hybrid optimization algorithm for nozzle structure. The results showed that the circular cross-sectional diameter, contraction angle, and divergence angle significantly affected the cavitation effect of the nozzle, while the inlet diameter had a smaller impact. Cui et al. [22] studied the effects of minor geometric deviations in nozzle internal shapes on cavitation and flow field conditions, showing that small differences in geometric structures still have a noticeable impact on cavitation flow characteristics. Han et al. [23] simulated submerged cavitation water jets generated by contraction-divergence nozzles using Fluent software and analyzed the impact of nozzle throat diameter and nozzle divergence angle on cavitation behavior in the flow field. The study found that increasing the nozzle throat diameter or decreasing the nozzle divergence angle would increase the overall vapor volume fraction in the flow field, with the highest impingement pressure measured experimentally reaching 300 MPa.

In summary, for traditional conical convergent nozzles, the acceleration time of the abrasive water jet is relatively short, and the strengthening effect is not ideal. To meet various complex service conditions, this study, based on numerical analysis and simulation, designed contraction-divergence nozzles with different divergence angles and derived functional relationships between divergence angles and nozzle exit velocities. The impact of divergence angles on the flow field structure was analyzed using Fluent finite element software, and the numerical results were validated. Finally, a nozzle structure suitable for surface strengthening of metals was proposed, along with the optimal nozzle divergence angle.

2. Nozzle design theory

2.1. Design principles of high-pressure abrasive water jet contraction-divergence nozzles

The contraction-divergence nozzle, also known as a Laval nozzle, has a simple structure with a tapered bore, where the diameter decreases initially and then increases. The smallest diameter segment is the cylindrical section. As shown in Fig. 1, this figure illustrates the schematic structure of a contraction-divergence nozzle. Its structure primarily comprises a convergent section, a cylindrical section, and a divergent section. In the process of abrasive water jet flow, the fluid medium accelerates through the convergent section and experiences laminar flow in the cylindrical section before entering the divergent section of the nozzle at extremely high speeds. The compression and motion of high-pressure abrasive water jets are determined by the length and

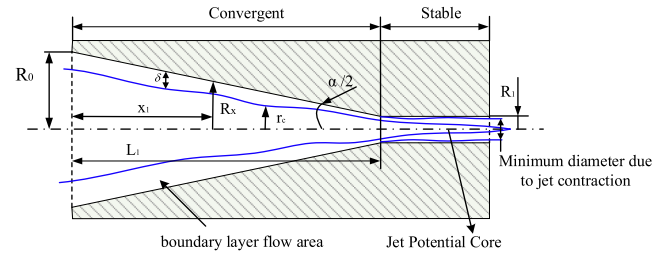


Fig. 2. Schematic diagram of the internal flow analysis of the converging section of the nozzle.

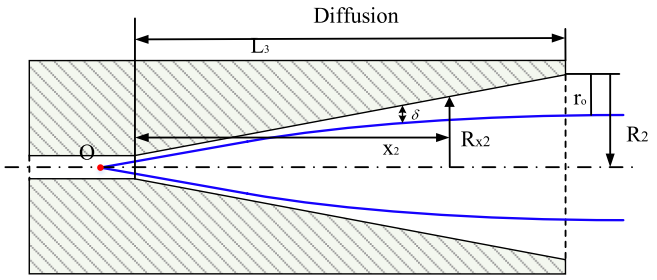


Fig. 3. Schematic diagram of the internal flow analysis of the diffusion section of the nozzle.

convergence angle of the convergent section [24–26]. After passing through the cylindrical section, the fluid medium enters the divergent section with even higher velocity, reaching its maximum at the nozzle exit. The divergence angle plays a crucial role in the exit pressure and velocity.

In the design process, the following assumptions are made:Under high pressure conditions (10–100 MPa), the fluid inside the contraction-divergence nozzle is treated as incompressible.The water jet at the inlet is assumed to be turbulent, and there is mass and momentum transfer between the air and water flow.

(1) Flow Coefficient

The flow coefficient of the nozzle characterizes its energy transfer capacity, and it is related to the nozzle's geometric structure and the flow state within the nozzle [27]. For a contraction-divergence nozzle, the flow coefficient Cd is defined as:

$$C_d = 1 - 4 \times \delta^* / d_0 \tag{1}$$

d_0-the nozzle exit diameter
δ^*-Boundary layer displacement thickness

(2) Layer Flow Transition

The phenomenon of layer flow transition was discovered by Narasimha and Sreenivansan in 1973 [28]. Layer flow transition refers to the occurrence of characteristics resembling laminar flow in a turbulent boundary layer region under high-speed conditions.

A straightforward and practical method for determining 2-layer flow transition is the pressure gradient parameter, denoted as Kp, which can be defined as:

$$K_P = \frac{\mu}{U}\frac{du}{dx} \tag{2}$$

μ-Dynamic viscosity of water.
U-Fluid velocity inside the nozzle.

When Kp is less than 0.5×10^{-6}, it is determined to be turbulent. When Kp is in the range of 0.5×10^{-6} to 3.0×10^{-6}, it is considered a transition from turbulence to laminar flow. When Kp is greater than 3.0×10^{-6}, it is considered a laminar boundary layer.

Research has shown that turbulent boundary layers are unfavorable for the density of the jet [29]. Therefore, the transition to laminar flow within the boundary layer is essential for the hydrodynamic performance of the nozzle. When designing water jet nozzles, this transition should be utilized to its full advantage.

(3) Boundary Layer Separation

In the fluid flow within the boundary layer region, when the velocity gradient at a certain point equals zero, boundary layer separation may occur. Boundary layer separation can occur on the inner wall of the nozzle inlet and in the outlet section. Boundary layer separation can lead to flow instability, thereby affecting the hydrodynamic performance of the jet. Therefore, efforts should be made to minimize it.

(4) Nozzle Flow Passage Wall Roughness

The degree of boundary layer turbulence in the nozzle is closely related to the nozzle's contour shape and dimensions. Simultaneously, the influence of wall roughness on the inner wall of the nozzle flow passage is also significant. Wang's experimental studies have shown that increasing the roughness of the inner wall of the nozzle leads to an increase in boundary layer thickness [30]. On the other hand, as the shape factor increases with roughness, the likelihood of boundary layer separation also increases.

To reduce turbulence levels, the roughness of the inner wall of the nozzle flow passage should be less than the thickness of the laminar sublayer at which the boundary layer separates (Sc), which is expressed as:

$$\Delta < S_c$$

This is a surface roughness requirement that must be met in the design process of the nozzle flow passage.

According to the theory of boundary layer flow, the thickness of the laminar sublayer of the boundary layer (Sc) is calculated as:

$$S_c = \frac{5\sqrt{2\nu}}{U\sqrt{C_f}} \tag{3}$$

The above analysis provides a theoretical foundation for the design and performance improvement of contraction-divergence nozzles.

2.2. Convergent section

The convergent section is located at the beginning stage of the nozzle structure and is the region where the internal jet acceleration within the entire nozzle has the greatest effect. An appropriate convergence angle can ensure the provision of a favorable acceleration environment while minimizing the excessive energy loss caused by over-constricted cross-sections. Research by Liu and others [31] has shown that the size of the convergence angle has almost no significant impact on liquid acceleration performance.

As shown in Fig. 2, the length of the convergent section (Lp) is related to the half-cone angle (α) of this section, the inlet diameter (D0), and the diameter of the cylindrical section (D1). The radius of any section within the convergent section is given by:

$$L_1 = \cot\frac{\alpha}{2}(R_0 - R_1) \tag{4}$$

$$R_x = R_0 - \int_0^{x_1} \tan\frac{\alpha}{2} dx \tag{5}$$

In the equation, L_1 is the length of the convergent section.R_0 is the

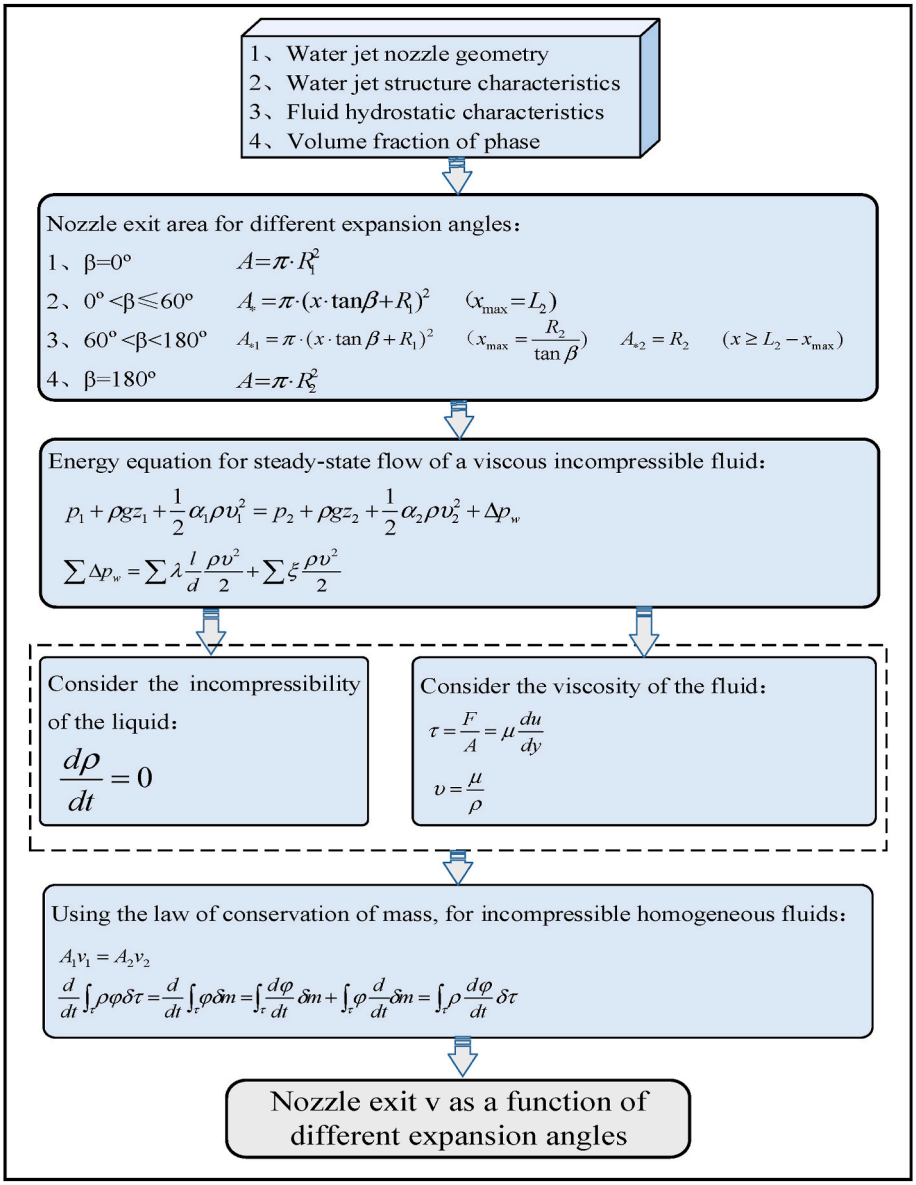


Fig. 4. Nozzle exit velocity flow chart.

half-radius at the nozzle inlet.R_1 is the half-radius at the exit of the convergent section (also the radius of the cylindrical section).α is the convergence angle of the convergent section.R_X is the radius at any axial distance X.

2.3. Cylindrical section

The cylindrical section of the nozzle is the intermediate (or final) stage through which the jet passes in the nozzle structure. Its primary function is to further stabilize the high-turbulence water jet and improve the jet's focus, allowing the impact of the jet to be concentrated [32]. Numerous experiments have shown that for conical nozzles, there exists an optimal range for the convergence angle (α), typically considered to be in the range of 8°–24°. It is generally believed that there exists an optimal dimensionless length of the cylindrical section (L2) for conical nozzles in the range of 2–4, which is the ratio of the length of the cylindrical section to the diameter of the cylindrical section (R_1).

2.4. Divergent section

The divergent section is a low-pressure region, and its purpose is to generate supersonic jet flow at the nozzle exit for a contraction-divergence nozzle [33]. Given the diameter of the cylindrical section (R_1), the length of the divergent section(L_3), and the divergence angle (β), the nozzle exit diameter is calculated as:

$$R_2 = 2L_3 \tan\frac{\beta}{2} + R_1 \tag{6}$$

2.5. Nozzle exit diameter

For continuous jet flow, applying the Bernoulli equation between the inner and outer points on the cross-section at the nozzle exit, while neglecting the height difference between the two points, we have:

$$p_\varepsilon + \frac{\rho_\varepsilon v_\varepsilon^2}{2} = p_\delta + \frac{\rho_\delta v_\delta^2}{2} \tag{7}$$

p_ε and p_δ-the internal and external static pressure;

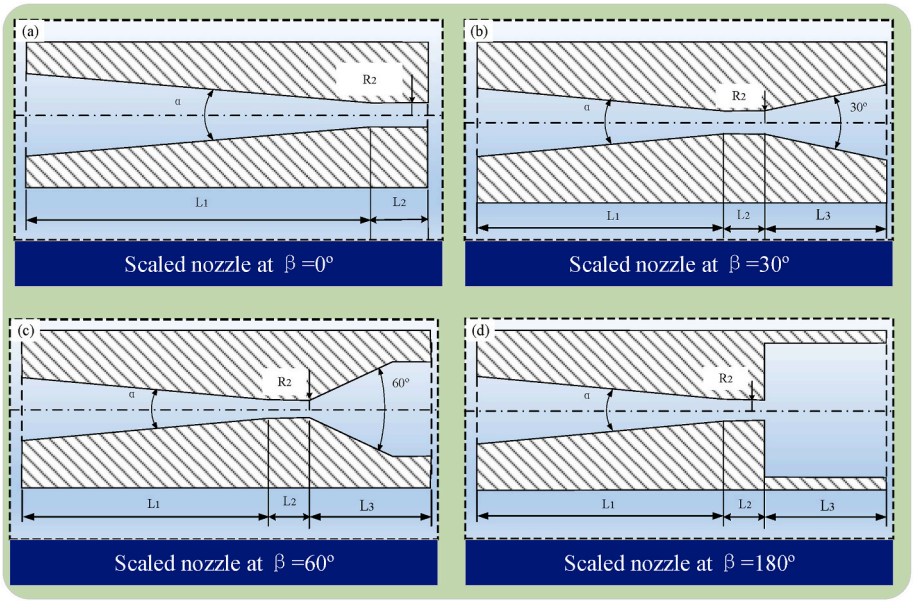


Fig. 5. Water jet nozzle structure with different extension angles:(a)β is 0°; (b)β is 30°; (c)β is 60°; (d)β is 180°.

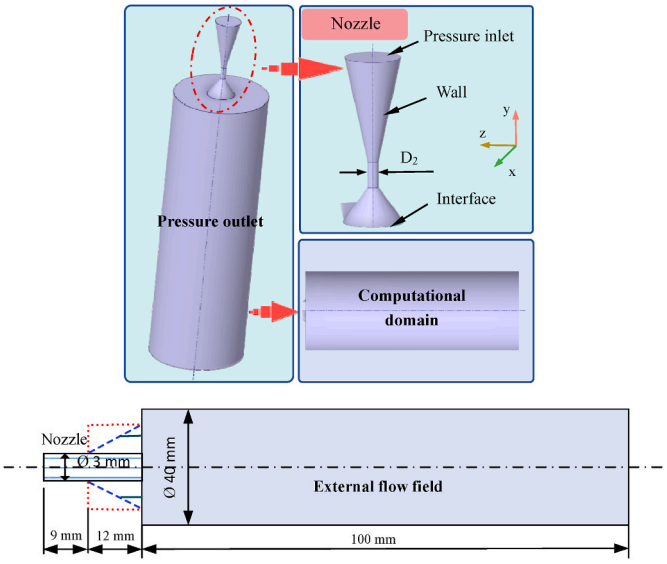


Fig. 6. Physical model for nozzle motion conditions.

v_ε and v_δ-The the internal and external average velocity;

Because the interior of the nozzle is a cylindrical structure, the area at any position $A = \pi\frac{d_x^2}{4}$, in conjunction with the law of mass conservation and formula (7), can be derived as follows:

$$v_\varepsilon = \sqrt{\frac{2(p_\delta - p_\varepsilon)}{\left[\rho_\varepsilon - \rho_\delta\left(\frac{d_\varepsilon}{d_\delta}\right)^4\right]}} \tag{8}$$

In the process of the water jet, due to $p_\delta \gg p_\varepsilon$, $d_\varepsilon \gg d_\delta$, and the density of water being 1000 kg/m^3, when substituted into the above equation, a simplified expression for the jet flow velocity can be obtained as follows:

$$v_\varepsilon = 44.7\sqrt{p_\alpha} \tag{9}$$

Based on the jet velocity $v = \frac{q}{A}$, the jet flow rate can be determined, where the jet flow rate is equal to the jet exit velocity multiplied by the cross-sectional area of the nozzle exit, resulting in $q_\alpha = 2.1d_\alpha\sqrt{p_\alpha}$:

The actual jet velocity and flow rate through the nozzle are typically less than the theoretical values. The ratio of the actual flow rate to the theoretical flow rate is defined as the flow coefficient, which can be expressed as:

$$Q_\beta = C_v \cdot Q_\beta = 2.1d_\alpha\sqrt{p_\alpha} \tag{10}$$

Simplifying the above equation:

$$d_\alpha = \sqrt{\frac{C_v Q_\alpha}{\sqrt{p_\alpha}}} \tag{11}$$

3. Construction of a prediction model for the nozzle exit velocity of water jets

In order to elucidate the flow state of the fluid medium inside nozzles with different divergence angles and considering the influence of dynamic viscosity, incompressibility, and external forces at the inlet, a model of the velocity at the nozzle exit section of water jets was developed (see Fig. 3). The flowchart of this method is depicted in Fig. 4.

3.1. Numerical simulation of water jet

Conical nozzles are commonly used as water jet nozzles and are generally considered a typical device for generating continuous jets. They convert the pressure potential energy generated by the nozzle pressure drop into the kinetic energy of the fluid medium (water), resulting in a high-speed jet at the nozzle exit, thus achieving surface strengthening of metal materials. Wen et al. [34] conducted research on a straight conical nozzle and found that the impact force of the water jet increases with the increasing exit diameter. A cylindrical exit section in straight conical nozzles is advantageous for improving the hydraulic performance of the water jet. Conical nozzles have a simple structure, and the continuous jet they produce is a process of continuous axial velocity decay. The key factors affecting this process are the properties of the fluid medium and the structural parameters of the conical nozzle. In this study, both the jet medium and the surrounding fluid medium are water, so the fluid medium is already determined, and the variable parameters are the structural parameters of the conical nozzle. Based on the design principles mentioned earlier and previous research [35–37], improvements were made to the typical conical nozzle. As shown in

Table 1
Structure parameters of nozzle.

Project	Parameter
Convergence angle	10 deg
Length of convergence section	18 mm
Diameter of cylindrical section	3 mm
Length of cylindrical section	9 mm
Length of divergent section	12 mm

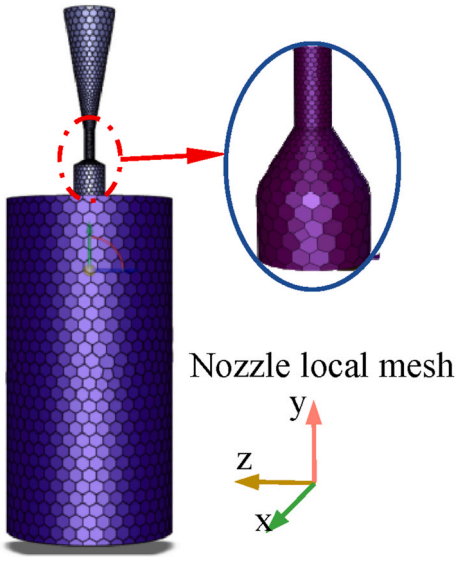


Fig. 7. Nozzle and flow field meshing.

Table 2
Water jet nozzle meshing parameters.

Diffusion angle	Nodes/pcs	Number of grids/pcs
0	50099	264199
30	154677	812458
60	164058	874810
90	171256	893544
120	179684	912547
180	181205	940272

Table 3
Nozzle diffusion section diffusion angle β and the relationship between the nozzle exit velocity.

β/°	30	60	90	120	180
Outlet speed/(m/s)	177	171	183	176	181

Fig. 4, several different water jet nozzle structures with varying divergence angles were designed. Among these, a nozzle with a divergence angle of 0° is the typical convergent nozzle, while angles from 0° to 30° are referred to as angled nozzles, angles from 30° to 60° are characterized by an angled nozzle with a straight section, and a divergence angle of 180° represents an abrupt divergence nozzle.

From Fig. 4, it can be observed that when the divergence angle is between 30° and 180°, there are cylindrical sections of varying lengths in the divergence section. Based on the designed convergent-divergent nozzle structure, when changing the divergence angle of the convergent-divergent nozzle, the relationship between the length of the divergence section and the nozzle exit radius is as follows:

$$\begin{cases} \beta = 0^{\circ} & R_3 = 0mm, L_3 = 0mm \\ 0^{\circ} < \beta < 30^{\circ} & R_x = x\ tan\dfrac{\beta}{2} + R_2, x_{max} = L_3 \\ 30^{\circ} < \beta \le 180^{\circ} & \begin{cases} R_{x1} = x_1\ tan\dfrac{\beta}{2} + R_2, x_{max} = \dfrac{R_3}{tan\beta/2} \\ x_2 = L_3 - x_{max} \end{cases} \end{cases} \tag{12}$$

For all fluids, they satisfy the basic equations of fluid mechanics [38] (continuity equation, motion equation, and energy equation). However, the fluid motion that satisfies the same set of equations can vary greatly. Therefore, only with specific initial conditions and boundary states, the fluid motion takes on a unique form. For the surface strengthening of high-pressure water jets in this paper, the set boundary conditions are as follows: the initial pressure at the inlet is set to be P_0.

During the operation of the water jet, under the action of a certain water pressure, the fluid accelerates through the convergent section and cylindrical section before entering the divergent section. Due to the change in flow state, the jet exit section forms an expanding cross-section. The diameter of the divergent section cannot be directly measured, so it is necessary to first calibrate the cross-sectional contraction coefficient(C_C) at the nozzle exit. Combining this with the geometric dimensions of the nozzle (i.e., the nozzle exit radius), the area of jet expansion at the exit section can be determined:

$$A = C_c A_d = \pi R^2 C_c \tag{13}$$

It can be seen that the cross-sectional expansion coefficient of the nozzle depends not only on the geometric shape and size but also on the flow state of the nozzle. At the nozzle inlet, applying the Bernoulli equation yields the inlet velocity of the nozzle:

$$v_1 = C_V\sqrt{\frac{2p_0}{\rho}} \tag{14}$$

In practical engineering applications, the flow inside the pipeline is often simplified as one-dimensional flow. Without considering fluid compressibility, according to the law of mass conservation, we can obtain:

$$A_1 v_1 = A_2 v_2 \tag{15}$$

Combining equations (14) and (15), the velocity at the nozzle exit for a divergence angle between 0° and 30° can be obtained as:

$$v_{out} = \frac{C_V\sqrt{2\Delta p/\rho}\cdot\left(L_1\cdot\tan\frac{\alpha}{2} + R_2\right)^2}{\left(L_3\cdot\tan\frac{\beta}{2} + R_2\right)^2} \quad 0^{\circ} < \beta < 30^{\circ} \tag{16}$$

However, when the nozzle divergence angle increases to 30° or more, a straight section appears in the diverging segment, and the flow field structure and turbulent boundary in the diverging segment change due to the presence of the straight section. In the conical section of the nozzle, it still follows the variation pattern of equation (21), and the jet velocity in the straight section of the nozzle should be calculated separately.

For the conical part of the diverging segment, the jet velocity still follows the law of mass conservation, and according to equation (16), the jet velocity at the cross-section of the conical part of the diverging segment can be obtained:

$$v_1 = \frac{C_V\sqrt{2p_0/\rho}\cdot\left(L_1\cdot\tan\frac{\alpha}{2} + R_2\right)^2}{\left(L_x\tan\frac{\beta}{2} + R_2\right)^2} \tag{17}$$

According to the research by Zheng et al. [39], in the process of plane turbulence, the velocity distribution of the jet at various cross-sections

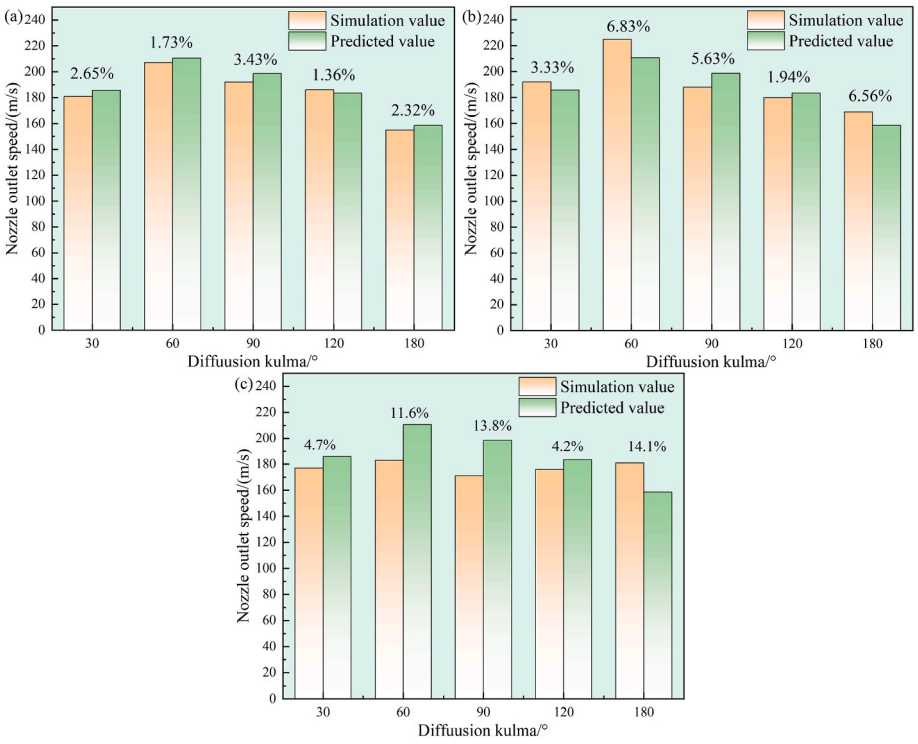


Fig. 8. Comparison of simulated and predicted values:(a)Size = 0.10 mm; (b)Size = 0.15 mm; (c)Size = 0.20 mm.

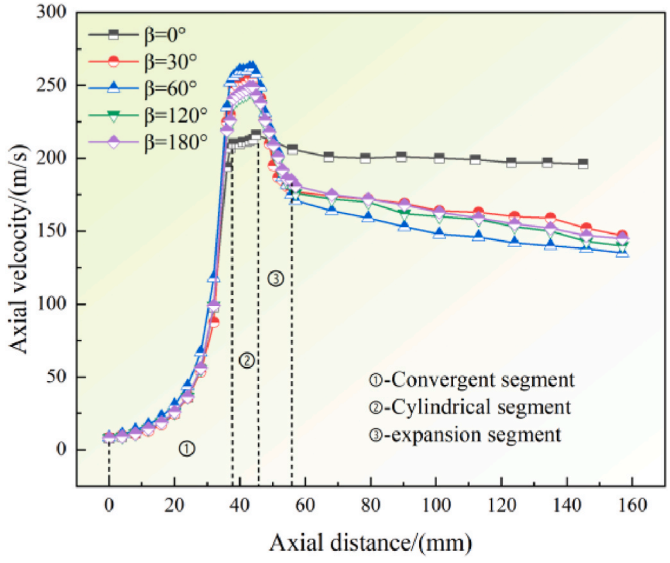


Fig. 9. The effect of diffusion angle on the axial velocity inside the nozzle.

exhibits similarity. The velocity is highest along the axis, decreases as one moves away from the axis, and comes to a stop at the nozzle wall. Here, v_y is defined as the velocity in the y-direction at any cross-section, while v_x is the velocity at the jet axis. The velocity distribution of the jet in the straight section of the nozzle can be considered as a Gaussian distribution:

$$\frac{v_y}{v_x} = \exp\left(-(x_2/b)^2\right) \tag{18}$$

From equation (10), the corresponding water jet flow rate can also be obtained as:

$$Q = Av_0 = \pi R^2 \varphi \sqrt{\frac{2P_0}{\rho}} \tag{19}$$

Furthermore, based on the fundamental equations, the kinetic energy of the jet can be obtained as E:

$$E = \frac{1}{2}\rho Q v_0^2 = \pi R^2 \varphi^3 \sqrt{\frac{2p_0^3}{\rho}} \tag{20}$$

It can also be observed from equations (14) and (19) that when the nozzle structure is determined, the jet's velocity is only related to the initial pressure of the nozzle. By increasing the initial pressure, the jet's velocity can be enhanced, and at the same time, as the initial pressure increases, the jet can gain more energy.

In the process of water jetting, the velocity on the axis is the highest, and its velocity gradually decreases radially outward. When neglecting the effects of gravity and friction losses on the jet, the momentum conservation law at each cross-section is:

$$\int_A \rho v_0^2 dA = \rho v_1^2 \pi R^2 \tag{21}$$

Substituting this equation into the momentum conservation law, we can obtain:

$$\int_0^x \frac{v_y}{v_x} \cdot \frac{x}{b} d\frac{x}{b} = 1 \Big/ 4\frac{v_0^2}{v_x^2}\frac{R^2}{b^2} \tag{22}$$

Considering the law of momentum flux conservation and the similarity of velocity distribution, together with the expression for the jet diffusion diameter, we can derive the velocity relationship for the straight section as:

$$\frac{v_x}{v_1} = \frac{a}{C\sqrt{2(e^2-1)}}\frac{R}{x} \tag{23}$$

Where: "a" is the natural constant and is taken as 2.718.

Finally, we get the jet outlet velocity for nozzles with a straight

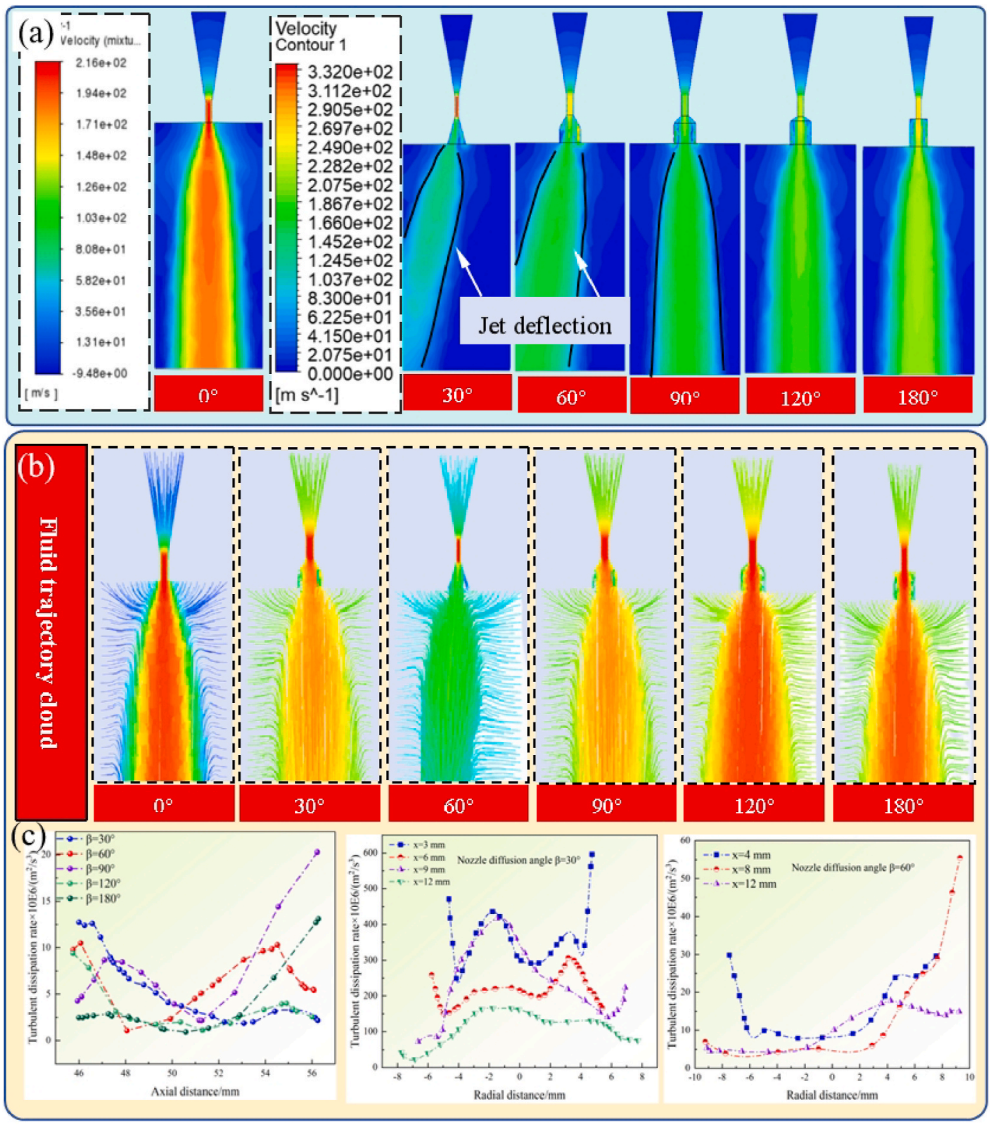


Fig. 10. Velocity simulation clouds, trajectory clouds and turbulent dissipation energy of nozzles with different diffusion angles:(a) Cloud map of velocity distribution inside nozzles with different diffusion angles; (b) The streamline trajectory of the internal flow field of the water jet; (c) The influence curve of diffusion angle on turbulent dissipation energy inside the nozzle.

section in the range of 30°–180° as:

$$v_{out}=\frac{aR}{Cx\sqrt{e^2-1}}\cdot\frac{C_V\sqrt{2p_0/\rho}\cdot\left(L_1\cdot\tan\frac{\alpha}{2}+R_2\right)^2}{\left(L_x\tan\frac{\beta}{2}+R_2\right)^2}\quad 30^\circ<\beta<180^\circ \tag{24}$$

4. Fluent nozzle outlet velocity simulation Verification

4.1. Governing equations and mathematical model

Since the internal flow in the nozzle is a complex turbulent flow, the Navier-Stokes (N-S) equations are chosen as the governing equations. However, the N-S equations, when combined with the fluid continuity equation and boundary condition equations, do not form a closed system. Therefore, additional conditions are needed to close the system. Hence, the standard k-ε turbulence model is employed. (1) In a spatial Cartesian coordinate system, the governing N-S equations for viscous, incompressible fluids can be expressed as:

$$\begin{cases} f_x-\frac{1}{\rho}\frac{\partial p}{\partial x}+\nu\left(\frac{\partial^2 v_x}{\partial x^2}+\frac{\partial^2 v_x}{\partial y^2}+\frac{\partial^2 v_x}{\partial z^2}\right)=\frac{dv_x}{dt} \\ f_y-\frac{1}{p}\frac{\partial p}{\partial y}+\nu\left(\frac{\partial^2 v_y}{\partial x^2}+\frac{\partial^2 v_y}{\partial y^2}+\frac{\partial^2 v_y}{\partial z^2}\right)=\frac{dv_y}{dt} \\ f_z-\frac{1}{p}\frac{\partial p}{\partial z}+\nu\left(\frac{\partial^2 v_z}{\partial x^2}+\frac{\partial^2 v_z}{\partial y^2}+\frac{\partial^2 v_z}{\partial z^2}\right)=\frac{dv_z}{dt} \end{cases} \tag{25}$$

f_x、f_y、f_z-The body forces per unit mass of fluid in the x, y, and z directions

This paper mainly analyzes and studies the nozzle flow process under non-submerged water jet conditions. Since nozzle flow is a free jet, considering the viscosity of water, it is assumed that the water flow is incompressible. In the spatial Cartesian coordinate system, for incompressible fluids, fluid elements can deform, but their volume remains constant. Thus, the continuity equation for water jet flow can be expressed as:

$$\frac{\partial(\rho v_x)}{\partial x}+\frac{\partial(\rho v_y)}{\partial y}+\frac{\partial(\rho v_y)}{\partial z}=0 \tag{26}$$

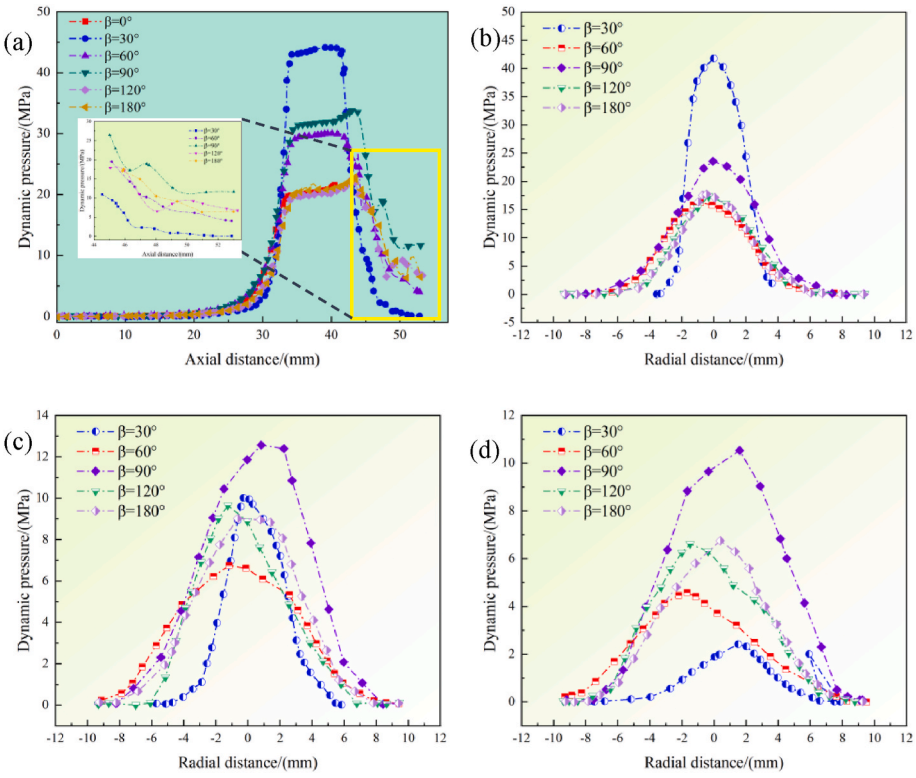


Fig. 11. Different diffusion angle nozzle axial dynamic pressure and longitudinal dynamic pressure distribution:(a)Axial distance; (b)Longitudinal distance of 31 mm; (c)Longitudinal distance of 35 mm(d)Longitudinal distance of 39 mm.

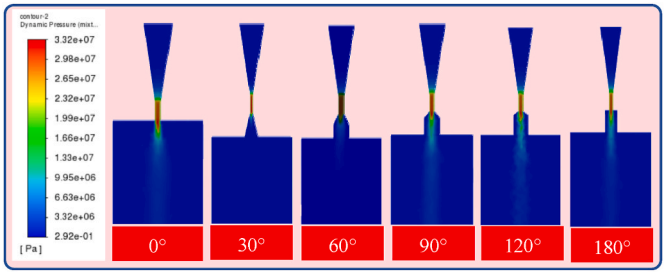


Fig. 12. Dynamic pressure simulation clouds for different diffusion angle nozzles.

(3) Standard k-ε Turbulence Model

The turbulence model used in this analysis is the standard k-ε two-equation model, where the turbulent viscosity can be expressed as a function of k and ε: The transport equation for turbulent kinetic energy k is given by:

$$\rho\frac{Dk}{Dt}=\frac{\partial}{\partial x_i}\left[\left(\mu+\frac{\mu_t}{\sigma_k}\right)\frac{\partial e}{\partial x_i}\right]+G_k+G_b-\rho e-Y_M(i=1,2) \tag{27}$$

$$\rho\frac{De}{Dt}=\frac{\partial}{\partial x_i}\left[\left(\mu+\frac{\mu_t}{\sigma_k}\right)\frac{\partial e}{\partial x_i}\right]+C_{1\varepsilon}\frac{e}{k}(G_k+C_{3e}G_b)-C_{2e}\rho\frac{e^2}{k}(i=1,2) \tag{28}$$

G_k-Turbulent kinetic energy caused by the average velocity gradient;
G_b-Turbulent kinetic energy caused by buoyancy effects;
Y_M-The impact of compressible turbulent pulsating expansion on the overall dissipation rate
μ_t-Turbulent viscosity coefficient.

Here $\mu_t=\rho C_\mu\frac{k^2}{e}$,$C_{1e}=1.44$、$C_{2e}=1.92$、$C_\mu=0.09$、$\sigma_k=1.0$ and $\sigma_e=1.3$.

4.2. Simulation model and parameters

Based on the design theory of the converging-diverging nozzle described above, the dimensions of the nozzle structure were defined. Other known parameters of the nozzle are listed in Table 1, and simulations were conducted using Fluent.

This paper focuses on the study of jet characteristics with water as the main medium. The Volume of Fluid (VOF) model was chosen since it is primarily used to track the interface position between two or more immiscible fluids. In this case, the two phases are air and water [40,41], where air is considered a compressible medium and water is incompressible. The simulations were based on the SIMPLE algorithm for pressure-velocity coupling. The relaxation factors for pressure, kinetic energy, turbulent kinetic energy (k), and dissipation energy (ε) were set to their default values, and the convergence criterion was set as the residual R ≤ 10^-6 during the iterative process.

In numerical simulations, the setting of boundary conditions is crucial for the accuracy and physical significance of simulation results [42,43].Boundary conditions not only determine the flow pattern of fluids in a specific region, but also directly affect parameters such as fluid velocity, pressure, and turbulence intensity [44].Reasonable boundary condition settings can ensure that the simulation model reflects the actual physical situation more realistically, making the simulation results more reliable and valuable for reference.

(1) Entrance boundary conditions

Pressure inlet conditions are set at the nozzle inlet. The inlet pressure p_0 is 20 MPa to achieve the turbulent state required for high-pressure water jet. The inlet pressure determines the velocity and kinetic energy of the fluid entering the nozzle, thereby affecting the jet

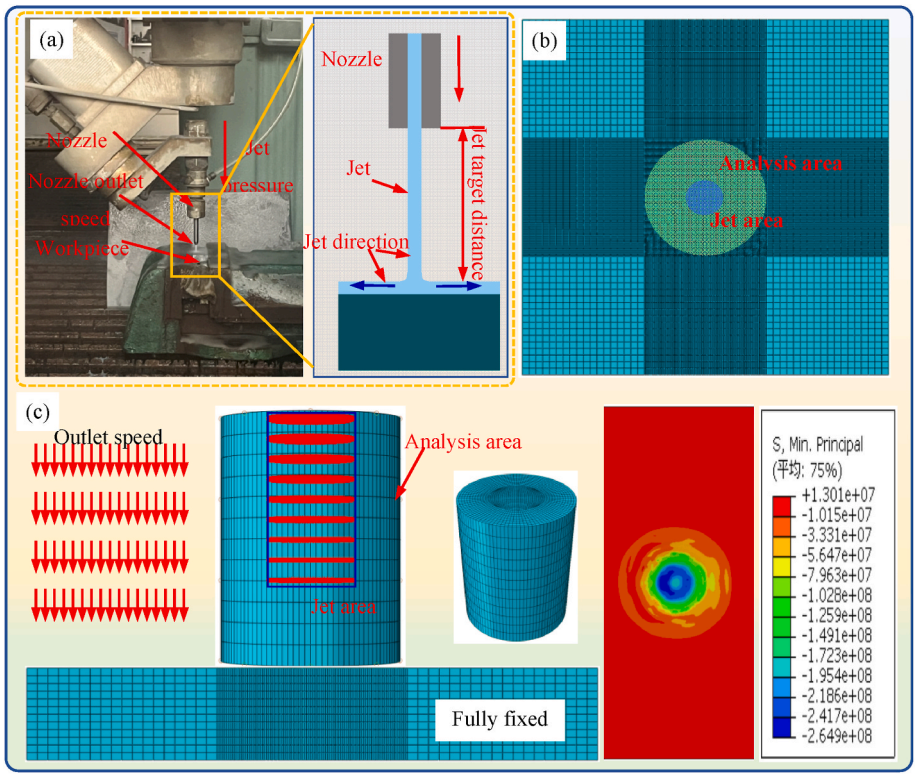


Fig. 13. Water jet simulation models.

Table 4
Material constants of 7075-T6 aluminum alloy.

A	*B*	*c*	*m*	*n*
527 MPa	692 MPa	0.017	1.61	0.71

Table 5
Diameter of jet beam for each diffusion angle.

Diffusion angle/°	Jet diameter
0	3 mm
30	7.93 mm
60	9.43 mm
180	9.43 mm

acceleration process.

$$v_{inlet} = \sqrt{\frac{2p_0}{\rho}} \tag{29}$$

In the equation, p_0 is the inlet pressure and ρ is the fluid density. In this simulation, the volume fraction of water is set to 0.9, so that the fluid composition is mainly water, to ensure the reproduction of cavitation phenomenon in the flow field and provide stable conditions for initial flow.

(2) Export boundary conditions

A pressure outlet is set at the outlet of the external flow field of the nozzle, with an outlet pressure of 0.1 MPa (atmospheric pressure), allowing the fluid to naturally diffuse after the nozzle outlet, which conforms to the free expansion characteristics of actual jets.The setting of the pressure outlet ensures that the jet can expand unconstrained at the outlet boundary, following the characteristics of free jet diffusion. According to the law of conservation of mass, it can be expressed as:

$$\rho v_{inlet} A_1 = \rho v_{out} A_2 \tag{30}$$

In the formula, A_1 and A_2 are the cross-sectional areas of the inlet and outlet, respectively. Through this condition, the expansion behavior of the fluid at the outlet can be accurately reflected.

(3) Wall boundary conditions

In order to simulate the friction and viscous effects on the inner wall of the nozzle, the wall is set as a non slip wall, meaning that the fluid velocity at the wall is zero, i.e. $v_{wall} = 0$.The non slip boundary condition ensures the frictional effect between the fluid and the inner wall of the nozzle, enabling a more realistic simulation of the turbulent structure and velocity gradient of the fluid boundary layer inside the nozzle. This is particularly important for capturing the flow characteristics of cavitation water jets in the boundary layer [45].At the same time, interface matching conditions are set between the inner wall of the nozzle and the contact surface of the flow field, ensuring the continuity of fluid velocity and pressure at the interface and ensuring the stability of boundary layer flow.

During the initialization of model parameters, air was assumed to fill both the interior and exterior flow fields of the nozzle, with 100 % air volume fraction throughout the computational domain. The interface between the nozzle and the computational domain was set to be matched with one another, allowing the transfer of jet energy across the interface. All other surfaces were treated as walls with no-slip boundary conditions. To enhance computational accuracy, double-precision calculations were employed, with a total of 1000 iterations, followed by solving the calculations. Throughout the simulation, parameters such as water jet dynamic pressure, total pressure, and the velocity distribution along the nozzle axis were used to analyze various jet characteristics.

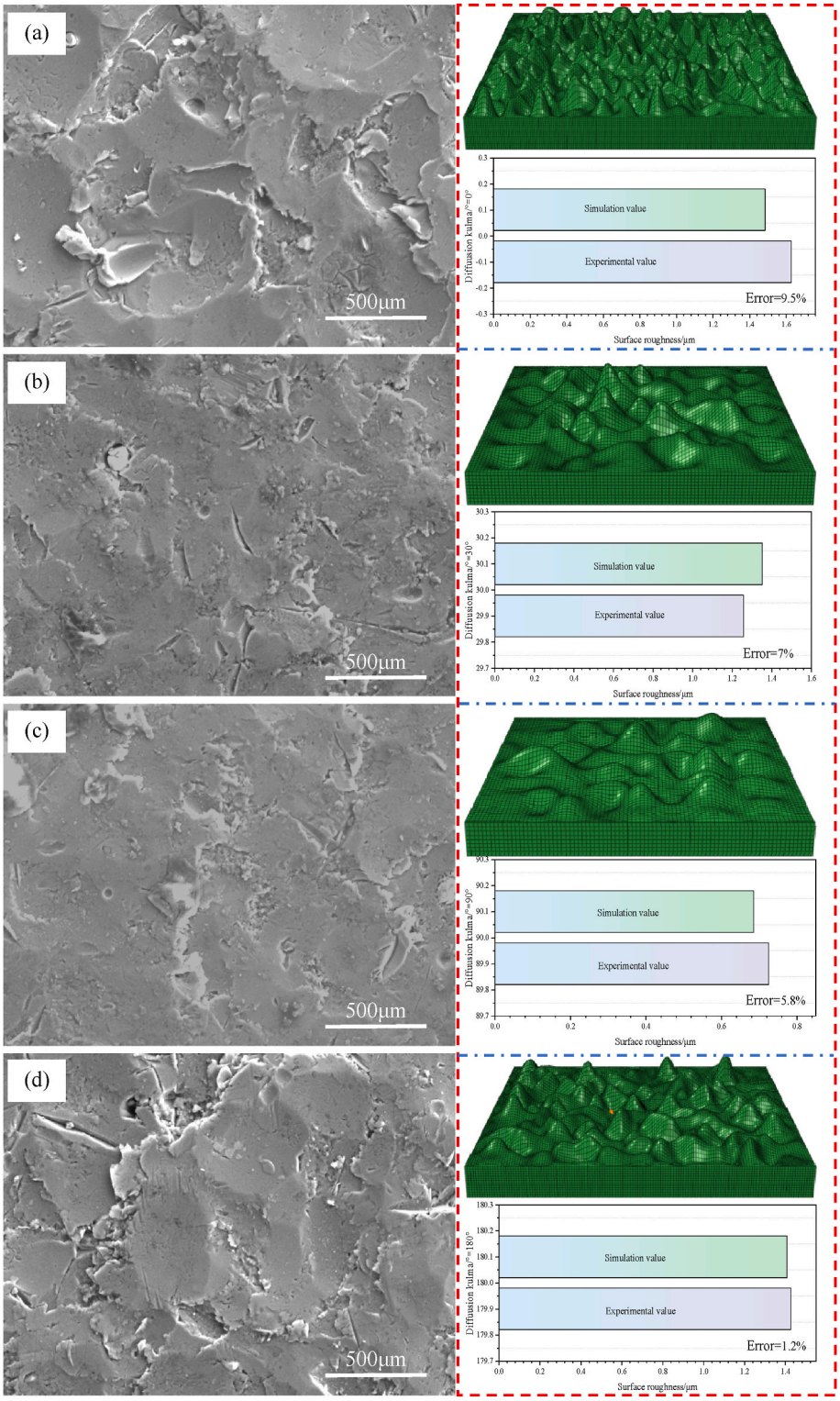


Fig. 14. The surface topography and roughness of 7075 aluminum alloy after impingement by water jets with varying diffusion angles:(a)0°; (b)30°; (c)90°; (d)180°.

4.3. Finite element mesh

In the research process, in order to accurately simulate the internal and external flow field structure of cavitation water jet nozzles, grid partitioning plays a crucial role in simulation accuracy and computational stability [46,47].Based on the geometric complexity of the nozzle and flow field, an unstructured mesh is adopted, which not only better adapts to the complex geometric details of the nozzle, but also helps to capture the characteristics of cavitation flow.In the highly turbulent and pressure changing areas inside the nozzle, a more refined grid layout was used to ensure the accuracy of simulation calculations. In addition, the narrow areas inside the nozzle and the high gradient areas of the flow field were locally densified to improve the grid quality of key areas.

The grid size of the key area is 0.1 mm to reduce potential numerical errors, while the grid size of the external flow field is 1 mm to optimize overall computational efficiency. The cavitation water jet grid

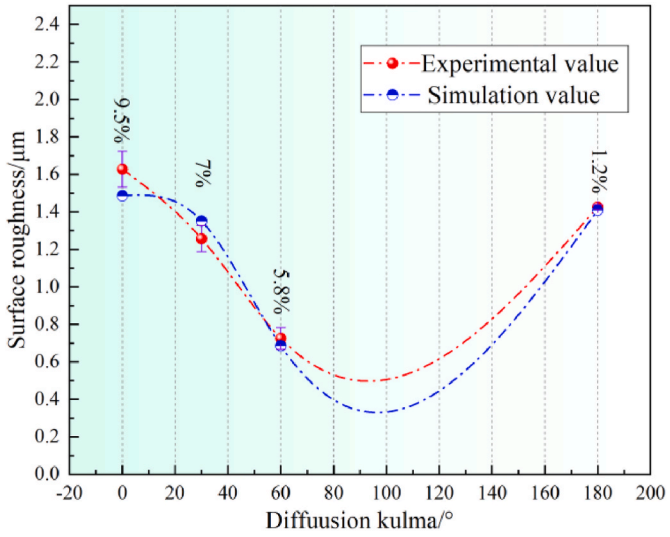


Fig. 15. Comparative study of experimental and simulated surface roughness for varying diffraction angles.

partitioning model is shown in Fig. 7.The grid of cavitation water jet nozzles under different diffusion angle conditions adopts the same partitioning strategy, and the specific grid parameters are shown in Table 2.

4.4. Results and analysis

(1) Validation of Nozzle Exit Velocity Prediction Model and Simulation Results

Table 3 shows the jet velocities at different nozzle exit angles obtained using Fluent finite element software. As observed in the table, the jet velocity at the exit increases with the increasing nozzle exit angle, showing an initial increase followed by a decrease. Based on the fitted exit velocity function relationships (Equations (16) and (24)) for different nozzle structures, the predicted nozzle exit velocities for various exit angles were obtained. This article compares the simulated values and predicted values of functional relationships based on different grid cells (Size = 0.10, 0.15, 0.20 mm), as shown in Fig. 8. According to the figure, when the grid size is small (Size = 0.10 mm) and there are more than 150000 grid cells, the simulated values are similar to the predicted values of the function, with an error between 1.36 % and 2.65 %.As the grid size increases, the maximum error is 14.1 %. This indicates that the established functional relationship has certain guiding significance for predicting the jet velocity at the nozzle outlet, and has a certain degree of accuracy, using a grid size of 0.1 mm.

(2) Impact of Divergence Angle on Axial Flow Velocity in the Nozzle

To validate the reliability of the functional relationship between inlet pressure and exit velocity mentioned earlier, a comparison was made for the axial velocity distribution of water jet nozzles with different divergence angles. Fig. 9 displays the axial velocity distribution curves inside the nozzle under various divergence angles. From the figure, it is evident that the increase in fluid medium velocity within the water jet nozzle primarily occurs in the convergent section, with a slight increase in jet velocity in the cylindrical section, while the jet velocity rapidly decreases in the divergent section. Due to the symmetric distribution of the water jet flow field, the velocity peaks at different axial distances are located on the nozzle's central axis. Moreover, it is noticeable from Fig. 9 that when the divergence angle $\beta = 0°$, the maximum nozzle exit velocity can reach 216 m/s. As the divergence angle increases, the nozzle exit velocity shows an increasing-decreasing-increasing trend. When the nozzle divergence angle is $\beta = 60°$, the nozzle exit velocity is the smallest, measuring 171 m/s. After entering the external flow field, the jet velocity exhibits a decreasing trend. The jet velocity at 60° shows the most significant speed attenuation, while the jet velocity at 0° experiences relatively minor variations in the initial section.

Analysis suggests that as the jet enters the convergent section, due to the small convergence angle and a long convergent section, there is significant resistance at the convergent section's inlet, which intensifies energy dissipation. According to the principle of energy conservation, fluid energy dissipation mainly reflects changes in static pressure [48]. The energy equation confirms that the rapid increase in jet velocity occurs in the convergent section. Once it enters the cylindrical section, the jet's turbulent intensity increases, and under the influence of gravity, the jet velocity increases accordingly. However, in the nozzle's divergent section, the high velocity gradient at the nozzle exit results in air being drawn into the nozzle, causing a contraction of the jet's profile at this location. As the jet stream enters the divergent section after acceleration in the cylindrical section, it induces high-speed shearing, leading to the immediate entrainment of surrounding fluid media, which decreases the velocity in the divergent section and reduces pressure. This consequently results in a decreasing trend in axial velocity. With an increasing nozzle divergence angle, it becomes less conducive to the formation of low-pressure areas, leading to an increase in the nozzle jet velocity. When the divergence angle increases beyond 30°, the divergent section includes a straight-line segment, causing differential entrainment in the divergent section, and consequently, the nozzle exit velocity exhibits an increasing-decreasing-increasing trend.

As shown in Fig. 10, it provides velocity and flow field distribution contour plots within the nozzle for different divergence angles. According to the figure, the flow velocity of the water jet inside the nozzle exhibits an increasing-decreasing trend, with more significant velocity changes when the nozzle divergence angle is small. When the nozzle divergence angle increases from 0° to 30°, it's evident that the jet experiences deflection in the external flow field. Furthermore, the farther the distance from the nozzle exit, the more pronounced the deflection. However, with a continuous increase in the nozzle divergence angle, the deflection phenomenon gradually disappears. As the fluid passes through the nozzle's contraction and cylindrical sections, it undergoes significant shearing and interpenetration with the air medium in the divergent section, resulting in extensive backflow phenomena near the walls. Once the water jet enters the jet flow field, it continues to interact with the surrounding fluid, causing a continuous reduction in velocity along the axis. Meanwhile, radially outward, the velocity decreases as you move further from the axis.

The analysis suggests that when the jet passes from the cylindrical section into the divergent section, the close proximity of the jet to the surrounding air allows for the exchange of energy, forming vortices. Fig. 10(b) also shows the streamline trajectories of the water jet, revealing that at divergence angles of 30° and 60°, the vortices formed in the divergent section are not symmetrical, which is the primary reason for the jet deflection. Xiao [49] found that the formation of vortices is closely related to turbulent dissipation energy. In Fig. 10(c), it can be observed that at a divergence angle of 30°, the dissipation energy decreases with axial distance, while at 60°, the dissipation energy exhibits a decreasing-increasing-decreasing trend with increasing axial distance. Simultaneously, in the radial distance, the energy dissipation is not symmetrical, being significantly higher on the right side of the axis compared to the left side. This corresponds to the streamline contour map in Fig. 10(b), indicating that in the jet process, turbulence is more intense on one side, consuming more energy and causing the jet to deflect without fully expanding.

As the water jet exits the nozzle's divergent section and moves further along the axial distance, its jet expansion diameter is calculated as follows:

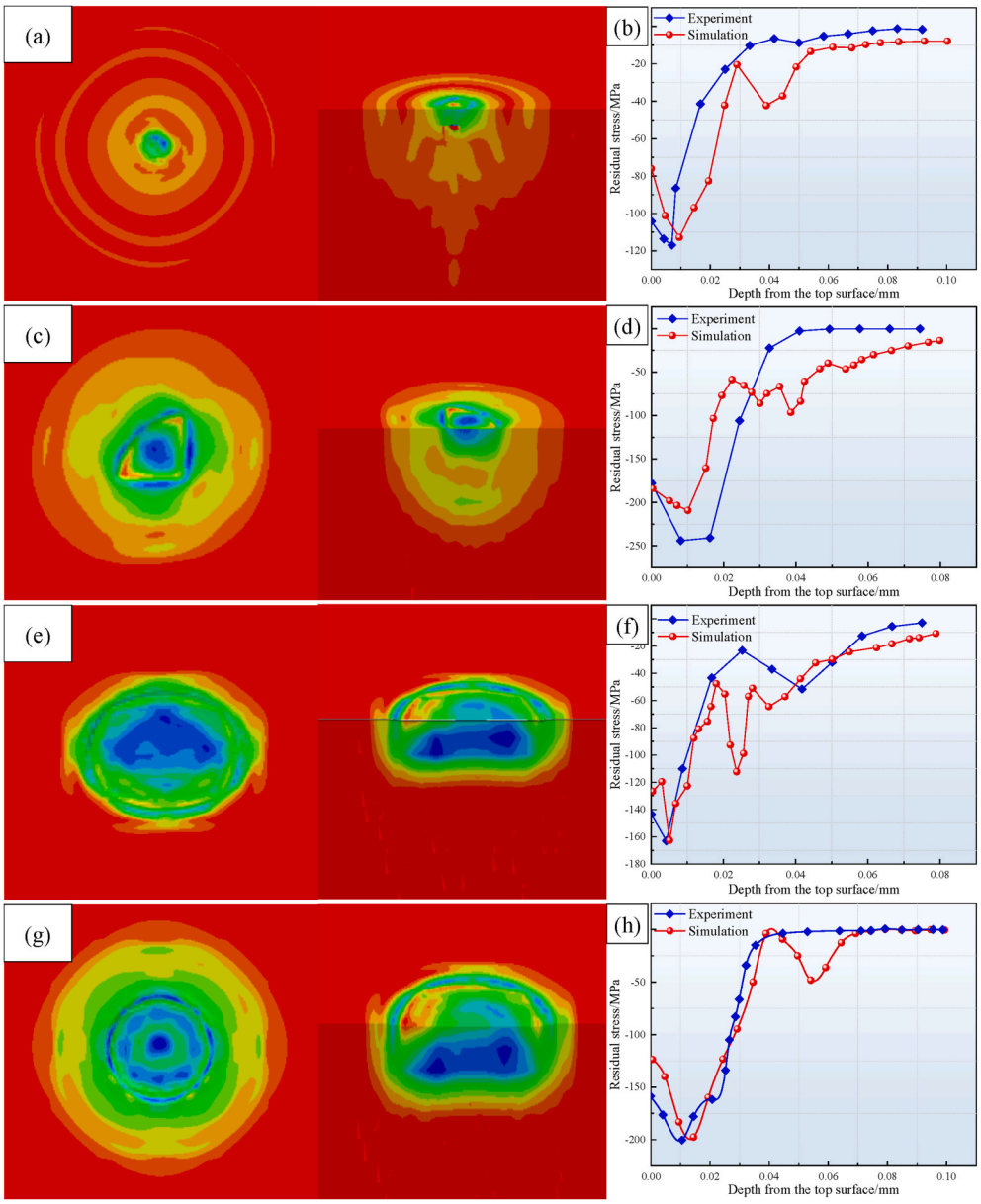


Fig. 16. Residual stresses and simulation clouds of nozzles with different diffusion angles:(a)(b)0°; (c)(d)30°; (e)(f)60°; (g)(h)180°.

$$b = 1 \Big/ 2d\left(\frac{C_d}{K_0}\right)^{1/2}\left(\frac{P_0}{P_x}\right)^{1/2} \tag{29}$$

In the equation you provided, here are the variables:

b: Jet expansion radius
C_d: Flow coefficient for the nozzle
K_0: Coefficient
p_0、 p_x: Initial dynamic pressure in the nozzle's divergent section
x: Distance from the exit of the cylindrical section

According to the law of momentum flux conservation, the initial momentum flux at the nozzle's exit section is equal to the momentum flux at any section of the jet, i.e.,

$$\frac{\pi}{4}d^2\rho v_0^2 = 2\pi\rho\int_0^b v_x^2 r dr \tag{30}$$

This indicates that when the nozzle's exit velocity is greater, the energy of the resulting jet is also higher, resulting in a stronger impact on the workpiece material.

(3) Impact of Diffusion Angle on Dynamic Pressure

Fig. 11 shows the axial and longitudinal dynamic pressure distribution for nozzles with different diffusion angles. It can be observed from the graph that with increasing axial distance, the jet dynamic pressure sharply increases in the convergent section, enters the cylindrical section, and then exhibits a decreasing trend in the divergent section.

The radial dynamic pressure distribution at different axial distances of the nozzle at 4 mm, 8 mm, and 12 mm from the exit of the cylindrical section is analyzed. The radial distance is selected in the z-direction according to Fig. 6.

From Fig. 11(b), (c), and (d), radial dynamic pressure distributions at various axial distances for nozzles with different diffusion angles can be observed. The overall trend is that the dynamic pressure is concentrated in the center of the jet, and it rapidly decreases radially from the geometric center. Pressure farther from the central axis of the water jet gradually decreases, approaching 0 MPa at the nozzle wall. The analysis

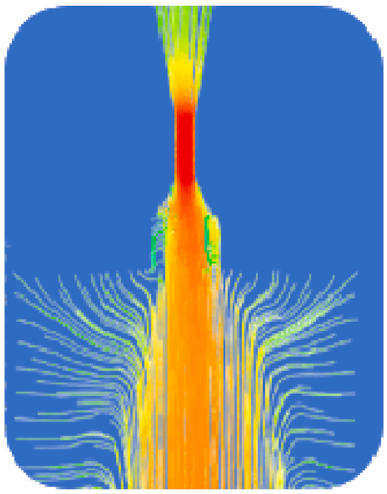


Fig. 17. Jet trajectory flow line diagram.

suggests that after leaving the cylindrical section, the water jet is affected by air resistance in the flow field, and its velocity continuously decreases, and it diverges radially. Dynamic pressure consequently decreases.

Combining the axial distance dynamic pressure distribution pattern in Fig. 11(a), it can be concluded that under the same working conditions, in the diffusion section of the water jet nozzle, the nozzle with $\beta = 90°$ has the highest peak outlet dynamic pressure and the best jet consolidation ability. In contrast, the nozzle with $\beta = 60°$ has the lowest dynamic pressure peak, as it experiences jet deflection in the diffusion section. When $\beta = 90°$, the water jet has the most significant divergence and the smallest central impact pressure. However, its impact range is more extensive, making it more suitable for metal surface strengthening operations. At the nozzle exit, dynamic pressure greater than 2 MPa is considered effective impact pressure. When $\beta = 60°$, the effective impact diameter is about six times larger than that of $\beta = 30°$, and it is 2–3 mm larger than other diffusion angles. This further indicates that the impact pressure and peak pressure of the water jet are related to the diffusion angle. In other words, the effective impact range is positively correlated with the diffusion angle, and the peak pressure shows a “W" trend with increasing diffusion angle.

Fig. 12 shows the dynamic pressure simulation cloud maps for nozzles with different diffusion angles. According to the graph, the dynamic pressure is most significant in the cylindrical section of the jet, and it increases with the diffusion angle. The dynamic pressure magnitude and axial distance exhibit a trend of first increasing and then decreasing in the divergent section, with the highest values observed at $\beta = 90°$.When a jet with a high Reynolds number is injected into a stationary environment, strong fluctuations occur at the interface between the jet and the surrounding stationary gas. The magnitude and frequency of these interface fluctuations differ as the jet progresses. When the jet enters the cylindrical section after the convergent section, the nozzle diameter sharply decreases, leading to intense turbulent fluctuations and higher dynamic pressure in the cylindrical section.The pressure field reveals that as the jet enters the divergent section, the diameter increases, resulting in increased air resistance and dissipation energy, leading to a gradual decrease in dynamic pressure. Additionally, with an increasing diffusion angle, the diameter gradually increases in the divergent section, causing the dynamic pressure to increase. However, when the diffusion angle reaches 60°, the initial large diameter in the divergent section intensifies air resistance, causing a decrease in dynamic pressure.

5. ABAQUS and experimental verification of water jet impact on target material

5.1. Experimental methods

To further validate the reliability of the fitted nozzle outlet velocity, experiments were conducted using 7075 aluminum alloy. Based on the nozzle outlet velocities obtained from Table 3, residual stresses on the target material after water jet impact were compared between the experiments and ABAQUS finite element simulations. The experimental equipment used was the HEAD42037Z water jet machine. The maximum output pressure is 420 MPa and the minimum output pressure is 10 MPa, as shown in Fig. 13 (a). The experimental setup is a jet pressure of 20 MPa and a jet target distance of 5 mm. After the experiment is completed, water jet nozzles with different diffusion angles are replaced.

Due to the instantaneous high temperature and high pressure process of water jet impacting the target material, elastic-plastic deformation is inevitable during this process. In order to better describe the elastic-plastic deformation of 7075 aluminum alloy during water jet impact [50],The Johnson Cook constitutive model is selected, and its expression is shown in equation (33). The material parameters of the Johnson Cook constitutive model for 7075 aluminum alloy are shown in Table 4 [51].

$$\overline{\sigma} = (A + B^n)\left(1 + C\ln\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right)\left[1 - \left(\frac{T - T_0}{T_m - T_0}\right)^m\right] \tag{33}$$

In the formula: σ is the equivalent stress; A is the yield strength; B is the hardening modulus; ε is the equivalent plastic strain; n is the strain hardening index; C is the strain rate sensitivity; $\dot{\varepsilon}$ is the strain rate; $\dot{\varepsilon}_0$ is the reference strain rate; T_0 is the reference ambient temperature (taken as room temperature); T_m is the melting point of the material; m is thermally sensitive.

When constructing the finite element model, the bottom of the workpiece is completely fixed with size parameters of $3 \times 6 \times 1$ mm and a jet target distance of 5 mm. According to Table 3, nozzle outlet velocities with different diffusion angles are set, and the nozzle outlet diameters (i.e. jet beams) with different diffusion angles are calculated based on Fig. 5, as shown in Table 5. At the same time, the impact area of the workpiece mesh was refined, and the simulation model is shown in Fig. 13.

5.2. Results and analysis

Fig. 14 shows the surface morphology and surface roughness of 7075 aluminum alloy after being impacted by water jets from nozzles with different diffusion angles. According to the figure, when the diffusion angle is 0°, there are significant pits on the surface of the workpiece after water jet impact, with a size of about 60 μm. As the diffusion angle increases to 30°, the surface quality of the workpiece improves, but there are still significant micro holes and cracks. When the diffusion angle is 90°, the surface roughness of the workpiece after water jet impact on 7075 aluminum alloy is the smallest.When the diffusion angle continues to increase from 90° to 180°, the surface roughness of the workpiece shows an increasing trend.Analysis suggests that when the nozzle diffusion angle is 30°, the jet deflects inside the nozzle, and the impact force spreads to a wider area and gradually loses its concentration, making the impact force insufficient to remove small particles and protrusions on the surface. On the contrary, the impact force on the surface becomes more scattered, resulting in a higher final surface roughness.As the diffusion angle increases to 90°, the jet beam has the best aggregation ability, and its impact force is concentrated in a certain area. Due to the large and concentrated impact force, it can effectively remove small particles and protrusions on the surface, thereby improving the surface roughness and making it smoother.

The experimental and simulation comparison curve of the surface roughness of the workpiece after water jet impact on 7075 aluminum

alloy is shown in Fig. 15. According to Fig. 15, the surface roughness shows a trend of first decreasing and then increasing with the increase of diffusion angle. The maximum error of the surface roughness measured by water jet impact on 7075 aluminum alloy and the surface roughness value obtained by simulation model is 9.5 %, and the minimum error is only 1.2 %. This indicates that the nozzle outlet velocity obtained according to the established functional relationship has certain reliability and accuracy.

Fig. 16 shows the residual stress and simulation cloud map after water jet impact on 7075 aluminum alloy for different diffusion angles. According to Fig. 16, the 7075 aluminum alloy exhibits mainly residual compressive stresses after water jet impact. The distribution of residual stress follows a "spoon" shape, where the maximum residual compressive stress is reached at a certain depth, after which it gradually decreases and approaches 0 with increasing depth. When the diffusion angle is 0°, the maximum residual compressive stress is 117 MPa. As the diffusion angle increases to 30°, the maximum residual compressive stress increases by approximately 133.3 MPa. With a continuous increase in the diffusion angle to 60°, the maximum residual stress decreases by 52.8 % compared to a 30-degree diffusion angle. However, when the diffusion angle is 180°, the maximum residual compressive stress reaches 201.6 MPa.

Comparing experimental values with simulation values, the changes in residual stress along the depth direction for different diffusion angles are generally similar. The maximum error is about 75 %, which occurs for a 60-degree diffusion angle at a depth of 22.8 μm. For the other diffusion angles and different depths, the errors in the experimental and simulation values range from 3.7 % to 39.6 %.

In Fig. 15, the simulation cloud map of residual stress also shows that the axial residual stress generated by water jet impact on 7075 aluminum alloy has a symmetrical distribution. The maximum axial residual compressive stress occurs just below the contact point of the jet with the workpiece. The analysis suggests that when a high-speed water jet acts on the workpiece surface, the jet velocity is converted into jet dynamic pressure. As seen in Fig. 11, the diffusion angle of 60° results in the smallest dynamic pressure peak, while a 30-degree diffusion angle leads to the largest dynamic pressure peak. This, in turn, results in more significant plastic deformation of the workpiece and the generation of substantial residual compressive stress at the impact point due to the opposing force created by the jet velocity.

5.3. Discussion

The numerical simulations and simulations show that changing the diverging angle of the convergent-divergent nozzle can improve the internal flow structure of the nozzle. Different nozzle diverging angles result in different jet velocities at the nozzle outlet. Simultaneously, alterations in nozzle structure are accompanied by flow field recirculation and entrainment effects. This paper focuses on discussing the impact of the deviation phenomenon in the internal flow field structure of the nozzle at different diverging angles on the nozzle outlet velocity.

By analyzing the nozzle outlet velocity at different diverging angles, as shown in Fig. 9, it is evident that the axial velocity of the jet significantly increases in the initial phase, remains stable in the cylindrical section, and then gradually decreases in the diverging section. Furthermore, when combined with the simulation cloud diagrams in Fig. 10, it can be observed that a diverging core region forms within the diverging section of the nozzle. Simultaneously, a deflection phenomenon occurs at diverging angles of 30° and 60°, with the jet's deviation occurring at the corner between the straight section and the conical section of the nozzle. As observed in Fig. 17, flow lines in this region undergo entrainment. Research by Zhou et al. suggests that jet entrainment is primarily due to the sudden hindrance of fluid motion in the entrainment boundary layer when it encounters lower-velocity fluid ahead [52]. As shown in Fig. 10(c), an increase in energy dissipation is evident at the corner. Due to jet deflection, the axial velocity is minimized when the diverging angle of the nozzle is 60°, at 171 m/s. Xiao et al.'s calculations indicate that when jet deflection occurs, the axial velocity reduction reaches 90 %. The length of the straight section increases with the larger diverging angle, with the dominant factor in jet focusing being the turbulent kinetic energy near the upstream region. The cross-sectional area between the outlet flow channel and the external air gradually increases, and the intensity of entrainment diminishes until it returns to focusing conditions.

In summary, it can be concluded that, under any jet pressure condition, the nozzle outlet velocity is minimized at a diverging angle of 60°, and the greater the diverging angle (i.e., the longer the straight section), the better the jet focusing performance. In general, lower jet velocities result in reduced pressure on the metal surface, leading to improved surface quality of the workpiece. Through the statistical analysis and comparisons presented, it becomes evident that the diverging angle can affect the nozzle outlet velocity, consequently reflecting its jet enhancement performance.

6. Conclusion

This study established an analytical model to predict the relationship between nozzle inlet pressure and nozzle outlet velocity for different diverging angles. Through Fluent simulations, the analysis of the internal flow field within the nozzle was conducted. The following conclusions were drawn:

(1) By comparing the predictive model to simulation results, it was observed that the errors within the range of diverging angles from 0° to 180° were approximately 10 %, which is within an acceptable range. This indicates that the model has some practical utility in predicting nozzle outlet velocity.

(2) The axial velocity in the nozzle experiences a sharp increase in the convergent section, remains relatively stable in the cylindrical section, and then dramatically decreases upon entering the divergent section. When the diverging angle is 0°, the nozzle outlet velocity can reach a maximum of 216 m/s. As the diverging angle increases, the nozzle outlet velocity follows a pattern of increase, decrease, and then increase again. The nozzle outlet velocity is the lowest at 60° diverging angle, measuring 171 m/s. Moreover, velocity simulation cloud diagrams show that at diverging angles of 30° and 60°, the jet exhibits deflection, and both the radial and axial velocities decrease continuously.

(3) The maximum dynamic pressure within the nozzle primarily concentrates in the cylindrical section and exhibits symmetric distribution. The maximum dynamic pressure along the jet direction is mainly located on the jet's axis, and it rapidly diminishes radially away from the geometric center. The effective impact range of the jet is greater at a diverging angle of 60° compared to other diverging angles.

(4) When the water jet impacts a 7075 aluminum alloy workpiece, the residual stresses are primarily residual pressure stresses. Both simulated and experimental residual stresses display a "spoon" pattern characterized by an initial decrease, followed by an increase, and eventually approaching zero. For a diverging angle of 30°, the maximum residual pressure stress on the 7075 aluminum alloy is 250.3 MPa. The residual stress values have errors ranging from 3.7 % to 39.6 %, demonstrating the reliability of the fitted nozzle outlet velocity model. Axial residual stresses generated by the jet impact on the 7075 aluminum alloy exhibit a symmetric distribution, with the maximum axial residual pressure stress occurring below the contact point of the jet with the workpiece.

CRediT authorship contribution statement

Xiujie Yue: Writing – original draft, Investigation. **Ping Zhang:** Writing – review & editing, Writing – original draft, Investigation, Funding acquisition. **Yajie Sun:** Investigation. **Xiaomin Jiang:** Methodology. **Youqiang Wang:** Writing – review & editing.

Funding

This study was financially supported by various sources, notably the National Natural Science Foundation of China (Grant No. 51705270), the Natural Science Foundation of Shandong Province (No. ZR2022QE149) .

Declaration of competing interest

The authors declare that they have no conflict of interest.

Data availability

No data was used for the research described in the article.

References

[1] Balaganesan Hariprasad, S. Joyson Selvakumar, D. Samuel Raj, Effect of cutting edge radius on end milling Ti–6Al–4V under minimum quantity cooling lubrication – chip morphology and surface integrity study, Wear (prepublish) (2022) 1–6.
[2] V.V. Kalyan Chakravarthy, T. Rajmohan, D. Vijayan, et al., Sustainable drilling of nano SiC reinforced Al matrix composites using MQL and cryogenic cooling for achieving the better surface integrity, Silicon (2021) 1–6 (prepublish).
[3] F. Müller, J. Monaghan, Non-conventional machining of particle reinforced metal matrix composite, Int. J. Mach. Tool Manufact. 40 (9) (2000) 1351–1366.
[4] Qingyun Fu, Chang Wang, Chengcheng Wu, et al., Investigating the combined effects of wide stacking faults and grain size on the mechanical properties and corrosion resistance of high-purity Mg, J. Alloys Compd. 927 (2022) 1–6.
[5] Dazheng Zhang, Xiuhua Gao, Mingming Wang, et al., Fatigue fracture behavior of high-strength low-alloy steel for flexible marine riser in the high cycle fatigue regime, Mater. Res. Express 9 (7) (2022) 1–6.
[6] Kashif Ali Abro, A fractional and analytic investigation of thermo-diffusion process on free convection flow: an application to surface modification technology, Euro. Phys. J. Plus 135 (1) (2020) 1583–15955.
[7] Yoshimura Toshihiko, Sugae Yuji, Ogi Takayuki, et al., Development of energy intensive multifunction cavitation technology and its application to the surface modification of the Ni-based columnar crystal superalloy CM186LC, Heliyon 7 (12) (2021) e08572.
[8] A. Mazurkiewicz, J. Smolik, The innovative directions in development and implementations of hybrid technologies in surface engineering, Arch. Metall. Mater. 60 (3) (2015) 2161–2172.
[9] L. Xiaohui, L. Qiuping, Z. Zuti, et al., Research on the underwater noise radiation of high pressure water jet propulsion, Ocean Eng. (2021) 219108438.
[10] X. Songqiang, G. Zhaolong, R. Qingyang, et al., Numerical analysis on the flow field structure and deflection characteristics of water jets under nozzle moving conditions, Eng. Applic. Comput. Fluid Mech. 14 (1) (2020) 1279–1301.
[11] Y. Liu, J. Zhang, J. Wei, et al., Optimum structure of a laval nozzle for an abrasive air jet based on nozzle pressure ratio, Powder Technol. (2020) 364343–364362.
[12] K. Thongkaew, J. Wang, H.G. Yeoh, Impact characteristics and stagnation formation on a solid surface by a supersonic abrasive waterjet, Int. J. Extrem. Manuf. 1 (4) (2019), 045004-045004.
[13] S. Wu, M. Xu, L.D. Hung, et al., Effects of nozzle configuration on internal flow and primary jet breakup of flash boiling fuel sprays, Int. J. Heat Mass Tran. (2017) 110730–110738.
[14] Yong Liu, Jianping Wei, Ting Ren, Zhaohui Lu, Experimental study of flow field structure of interrupted pulsed water jet and breakage of hard rock, Int. J. Rock Mech. Min. Sci. 78 (2015) 253–261.
[15] Peizhuang Han, Yabin Gao, Fei Wang, Jie Ren, Hao Zheng, Study on the influence of contact surface curvature on water jet impingement flow field and initial damage of coal seam, Energy Sci. Eng. 10 (9) (2022) 3331–3343.
[16] Y. Liu, H. Jia, W. Zuo, et al., Methodology and performance of annular fluid enhanced self-excited oscillation in pulsed submerged waterjet for coal mine engineering, Powder Technol. (2024) 435119401.
[17] S. Hloch, M. Srivastava, A. Nag, et al., Effect of pressure of pulsating water jet moving along stair trajectory on erosion depth, surface morphology and microhardness, Wear 452–453 (prepublish) (2020) 203278, 203278.
[18] Z. Wenjin, L. Huasheng, Y. Xiaoyuan, et al., Uncovering the liquid phase sintering mechanism of binderless cemented carbides, Ceram. Int. 49 (7) (2023) 11623–11629.
[19] C. Gutiérrez-Montes, R. Bolaños-Jiménez, A. Sevilla, et al., Bubble formation in a planar water–air–water jet: effects of the nozzle geometry and the injection conditions, Int. J. Multiphas. Flow 65 (2014) 38–50.
[20] Y.C. Jie, Z.C. Shou, A prediction model of wall shear stress for ultra-high-pressure water-jet nozzle based on hybrid BP neural network, Eng. Applic. Comput. Fluid Mech. 16 (1) (2022) 1902–1920.
[21] Yanzhen Chen, Yihuai Hu, Shenglong Zhang, Structure optimization of submerged water jet cavitating nozzle with a hybrid algorithm, Eng. Applic. Comput. Fluid Mech. 13 (1) (2019) 591–608.
[22] Jiwen Cui, Houhu Lai, Kunpeng Feng, et al., Quantitative analysis of the minor deviations in nozzle internal geometry effect on the cavitating flow, Exp. Therm. Fluid Sci. 94 (2018) 89–98.
[23] Bing Han, Z.H.A.O. Gang, et al., Influence of nozzle geometric parameter on cavitation behavior in water-jet cavitation peening processing, Mech. Sci. Technol. Aerospace Eng. 35 (5) (2016) 711–716.
[24] Bang Boo-Hyoung, Yong-Il Kim, Seokgyu Jeong, et al., Theoretical model for swirling thin film flows inside nozzles with converging-diverging shapes, Appl. Math. Model. 76 (2019) 607–616.
[25] L. Zhang, Q. Yu, T. Liu, et al., Coupled modeling and numerical simulation of gas flows laden with solid particles in de Laval nozzles, Shock Waves (2021) 1–18 (prepublish).
[26] Qinghong Zhang, Zhouhao Shi, Weidong Shi, et al., Research on flow field characteristics in water jet nozzle and surface damage caused by target impact, Sustainability 14 (15) (2022) 9074, 9074.
[27] Dongjae Kong, Seung Jin Song, Influence of flow coefficient on unsteady impeller loading induced by impeller-diffuser interaction, J. Mech. Sci. Technol. 36 (5) (2022) 2403–2413.
[28] R. Narasimha, K.R. Sreenivasan, Relaminarization in highly accelerated turbulent boundary layers, J. Fluid Mech. 61 (3) (1973) 417–447.
[29] J.J. Nijdam, T.A.G. Langrish, D.F. Fletcher, Assessment of an Eulerian CFD model for prediction of dilute droplet dispersion in a turbulent jet, Appl. Math. Model. 32 (12) (2007) 2686–2705.
[30] Chao Wang, Peijuan Cao, Chunhui Li, et al., Influence of wall roughness on boundary layer transition position of the sonic nozzles, Measurement 139 (2019) 196–204.
[31] Yong Liu, Juan Zhang, Jianping Wei, et al., Optimum structure of a laval nozzle for an abrasive air jet based on nozzle pressure ratio, Powder Technol. 364 (C) (2020) 343–362.
[32] G.O.N.G. Zhaoxin, L.U. Chuanjing, L.I. Jie, et al., The gas jet behavior in submerged Laval nozzle flow, J. Hydrodyn. 29 (6) (2017) 1035–1043.
[33] Lite Zhang, Mengyu Su, Feng Zilong, et al., Numerical study on the shock vector control performance in a de Laval nozzle with single or dual injection ports, J. Mech. Sci. Technol. 36 (6) (2022) 3001–3016.
[34] Jiwei Wen, Ziwei Qi, Seyed Saleh Behbahani, et al., Research on the structures and hydraulic performances of the typical direct jet nozzles for water jet technology, J. Braz. Soc. Mech. Sci. Eng. 41 (12) (2019) 1–12.
[35] Tianwen Jiang, Huang Zhongwei, Li Jingbin, et al., Experimental investigation of internal and external flow fields of jetting nozzles with different structures, J. Petrol. Sci. Eng. 217 (2022).
[36] Yong Liu, Juan Zhang, Tao Zhang, et al., Optimal nozzle structure for an abrasive gas jet for rock breakage, Geofluids 2018 (2018) 1–14.
[37] Jiwei Wen, Chen Chen, Ziwei Qi, et al., Bionic optimum design of straight cone nozzle and the effectiveness evaluation of reducing fluid resistance, J. Braz. Soc. Mech. Sci. Eng. 41 (9) (2019) 1–21.
[38] Xu Longfei, Huayun Shen, Wei Junxia, Mathematical foundations of the three-dimensional neutron hydrodynamics coupled transport equations, Ann. Nucl. Energy 156 (2021) 1–6.
[39] Yabin Gao, Xiaoya Guo, Xin Xiang, et al., Analysis on flow field and shape structure of circular nozzle water jet, China Saf. Sci. J. 31 (10) (2021) 82–88.
[40] Zheng Hong Xiang, Yun Luo, Zang Jing Yu, et al., Effect of nozzle outlet type on the flow field velocity and impact pressure of high-pressure water jet peening, J. Fluid Eng. 144 (7) (2022) 1–6.
[41] Yanbao Liu, Quanbin Ba, Lipeng He, et al., Study on the rock-breaking effect of water jets generated by self-rotatory multinozzle drilling bit, Energy Sci. Eng. 8 (7) (2020) 2457–2470.
[42] W. Maoxing, Y.C. Meng, N.C. Sheng, Recent advances in understanding propeller jet flow and its impact on scour, Phys. Fluid. 32 (10) (2020).
[43] A.J. Moríñigo, J.J. Salvá, Robust non-reflecting boundary conditions for the simulation of rocket nozzle flow, Aero. Sci. Technol. 14 (6) (2010) 429–441.
[44] M. Gholamisheeri, S.I. Wichman, E. Toulson, A study of the turbulent jet flow field in a methane fueled turbulent jet ignition (TJI) system, Combust. Flame (2017) 183194–183206.
[45] A. G B, J. Peter, J. Vincent, et al., Importance of the nozzle-exit boundary-layer state in subsonic turbulent jets, J. Fluid Mech. (2018) 85183, 124.
[46] S. Niu, H. Tong, X. Liu, et al., A quick simulation workflow to optimizing natural ventilation for building and landscape design, Energy Build. (2024) 324114875, 114875.

[47] A. Ravanji, A. Lee, J. Mohammadpour, et al., Optimising shapes of multiple pin fins in a microchannel using deep reinforcement learning and mesh deformation techniques, Appl. Therm. Eng. (2024) 256124099, 124099.
[48] Omidreza Sadeghi, Marjan Bakhtiari-Nejad, Fatemeh Yazdandoost, et al., Dissipation of cavitation-induced shock waves energy through phase transformation in NiTi alloys, Int. J. Mech. Sci. 137 (2018) 304–314.
[49] Songqiang Xiao, Zhaolong Ge, Ren Qingyang, et al., Numerical analysis on the flow field structure and deflection characteristics of water jets under nozzle moving conditions, Eng. Applic. Comput. Fluid Mech. 14 (1) (2020) 1279–1301.
[50] W. Yafei, X. Jianrui, Z. Yuexin, et al., Tensile properties and a modified s-Johnson-Cook model for constitutive relationship of AA7075 sheets at cryogenic temperatures, J. Alloys Compd. (2023) 942.
[51] Kedar S. Pandya, Christian C. Roth, Dirk Mohr, Strain rate and temperature dependent fracture of aluminum alloy 7075: experiments and neural network modeling, Int. J. Plast. 135 (2020) (prepublish).
[52] Zhou Lei, Zhou Wu, Guo Yanang, et al., Experimental studies on coherent structures in jet entrainment boundary layers, CIE J. 20 (7) (2019) 2520–2527.